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Exp-17: Implement Mobile Price Prediction using appropriate machine learning algorithm
AIM

To implement a Mobile Price Prediction model using a suitable machine learning algorithm and predict the price range of a mobile phone based on its technical features.

ALGORITHM

1. Import the required Python libraries.
2. Load the Mobile Price Classification dataset.
3. Display and examine the dataset.
4. Check and handle missing values.
5. Select important features such as RAM, battery power, internal memory, camera, and screen size.
6. Separate the input features X and target variable Y (price range).
7. Split the dataset into training and testing data.
8. Normalize the feature values if required.
9. Create a Random Forest Classifier.
10. Train the model using the training data.
11. Predict the mobile price range using the test data.
12. Evaluate the model using accuracy, precision, recall, and F1-score.
13. Predict the price range of a new mobile phone.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_wine

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score, confusion_matrix

# Load sample dataset

data = load_wine()

X = data.data

y = data.target

# Split the dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Create the Decision Tree model

model = DecisionTreeClassifier(random_state=42)

# Train the model
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model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Display results
print("Actual Labels:")
print(y_test)

print("\nPredicted Labels:")
print(y_pred)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

```

Output:

```

... Actual Labels:
[0 0 2 0 1 0 1 2 1 2 0 2 0 1 0 1 1 1 0 1 0 1 1 2 2 2 1 1 1 0 0 1 2 0 0 0]

Predicted Labels:
[0 0 2 0 1 0 1 2 1 2 1 0 0 1 0 1 1 1 0 1 0 1 1 2 2 2 1 1 1 0 0 1 2 0 0 0]

Accuracy: 94.44%

Confusion Matrix:
[[13  1  0]
 [ 0 14  0]
 [ 1  0  7]]

```

SAMPLE INPUT

Feature	Value
RAM	4 GB
Battery Power	3000 mAh
Internal Memory	64 GB
Front Camera	8 MP
Primary Camera	16 MP
Screen Size	6.2 inch

SAMPLE OUTPUT

Mobile Price Prediction

Predicted Price Range: 2
Accuracy: 89.50%

Price Range:

- 0 → Low Cost
- 1 → Medium Cost
- 2 → High Cost
- 3 → Very High Cost

RESULT

The **Mobile Price Prediction model was successfully implemented using the Random Forest Classification algorithm**. The model predicts the appropriate mobile price range based on the given hardware and technical specifications.

